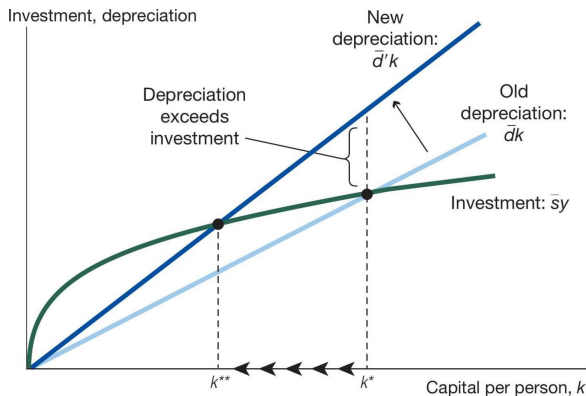


Macroeconomic Theory and Policy: Lecture 7 (Ch. 5)

A Rise in the Depreciation Rate

- Suppose the depreciation rate is higher rate ($\delta' > \delta$) due to an exogenous shock
 - ▶ The depreciation curve *rotates* upward
 - ▶ The investment curve remains unchanged
 - ▶ The capital stock declines until it reaches the new steady state
 - ★ This happens because depreciation exceeds investment ($\delta y_t > \delta' y_t$)
 - ▶ The new steady state is located to the left, k^{**}

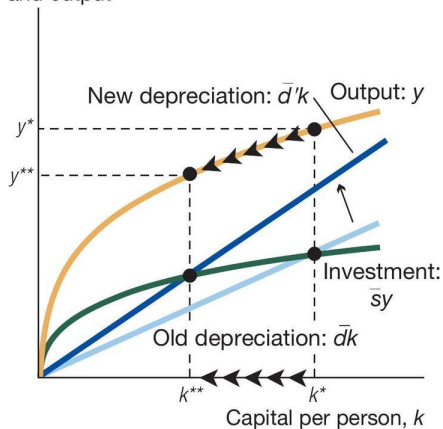


A Rise in the Depreciation Rate

- What happens to output in response to this increase in the depreciation rate?
 - ▶ The decline in capital reduces output
 - ▶ Output declines rapidly at first, and then gradually settles down at its new, lower steady state level y^{**}

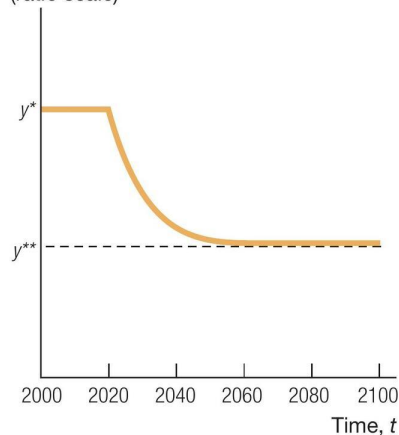
Behaviour of Output after an Increase in δ

Investment, depreciation,
and output



(a) The Solow diagram with output.

Output per person, y
(ratio scale)



(b) Output per person over time.

The Principle of Transition Dynamics

- If an economy is below steady state: It will grow
- If an economy is above steady state: Its growth rate will be negative
- The farther below its steady state an economy is (in percentage terms), the faster the economy will grow
- The closer to its steady state, the slower the economy will grow
- This allows us to understand why economies grow at different rates
 - ▶ Recall the convergence theory discussed earlier: poor countries tend to grow faster than developed countries
 - ▶ Based on Solow model, it could mean the poor countries are just further away from its steady state

Looking at Data through the Lens of the Solow Model

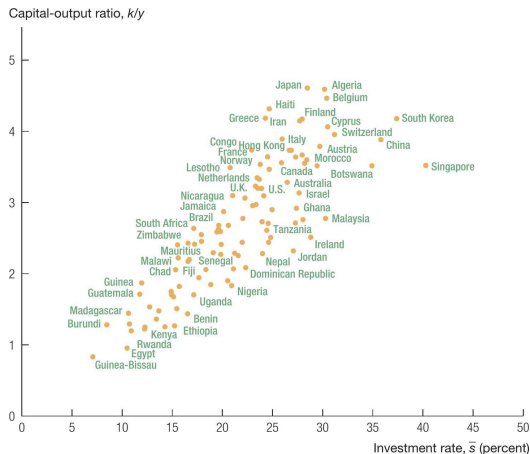
- Rearrange the steady state $\bar{s}y^* = \delta k^*$:

$$\frac{k^*}{y^*} = \frac{\bar{s}}{\delta}$$

- The capital to output ratio is the ratio of the investment rate to the depreciation rate
- Over the last 30 years,
 - ▶ The investment rate in South Korea has averaged more than 35 percent of GDP
 - ▶ The investment rate in the United States has been about 25 percent
 - ▶ In the poorest countries of the world, investment rates are only around 10 percent
- However, it is conventional to assume that the depreciation rate is relatively similar across countries, about 7 percent
 - ▶ if the δ is constant, then high investment rates should have high capital–output ratios

Explaining Capital in the Solow Model

- There is a strong positive relationship between the investment rate and capital-output ratio, as predicted in the model



Source: Penn World Table, Version 10.0. The capital-output ratio is measured in the year 2019, while the investment rate is averaged over the period 1990 to 2019.

Differences in Output per Capita

- The Solow model shows TFP is more important in explaining per capita output

- ▶ Let $\alpha = 1/3$, $\frac{1}{1-1/3} = 3/2$ and $\frac{1/3}{1-1/3} = 1/2$

$$y^* = \bar{A}^{\frac{1}{1-\alpha}} \left(\frac{\bar{s}}{\delta} \right)^{\frac{\alpha}{1-\alpha}} = \bar{A}^{3/2} \left(\frac{\bar{s}}{\delta} \right)^{1/2}$$

- ▶ Differences in capital are explained in part by differences in investment rate, and in part by differences in the productivity parameter
- We can calculate the significance of investment rates for explaining differences in y^* . Income per capita (y) is decomposable into
 - ▶ TFP differences
 - ▶ Investment differences

Differences in Output per Capita

- Take the ratio of y^* for two countries, assuming the depreciation rate is the same:

$$\frac{y_{rich}^*}{y_{poor}^*} = \left(\frac{\bar{A}_{rich}}{\bar{A}_{poor}} \right)^{3/2} \times \left(\frac{\bar{s}_{rich}}{\bar{s}_{poor}} \right)^{1/2}$$

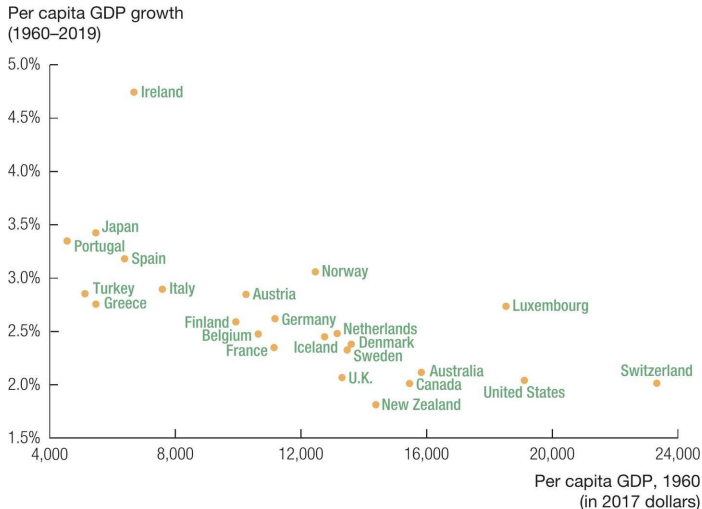
- Five richest countries to GDP in the five poorest countries, they differ by a factor of 64 ($\frac{y_{rich}^*}{y_{poor}^*} = 64$)
- In the richest countries, investment is about 25 to 30 percent. In the poorest countries, investment is about 7 percent. ($\left(\frac{\bar{s}_{rich}}{\bar{s}_{poor}} \right)^{1/2} = 4^{1/2} = 2$)
- Then, using the equation, $\left(\frac{\bar{A}_{rich}}{\bar{A}_{poor}} \right)^{3/2} = 32$

Understanding Differences in Growth Rates

- Empirically, for Organization for Economic Cooperation and Development (OECD) countries, transition dynamics holds:
 - ▶ Countries that were poor in 1960 grew quickly
 - ▶ Countries that were relatively rich grew slower
- For the world as a whole, on average, rich and poor countries grow at the same rate
 - ▶ Two implications of this:
 - ★ Most countries (rich and poor) have already reached their steady states
 - ★ Countries are poor not because of a bad shock, but because they have parameters that yield a lower steady state (determinants of the steady state investment rates and A)

Growth Rates in the OECD, 1960–2014

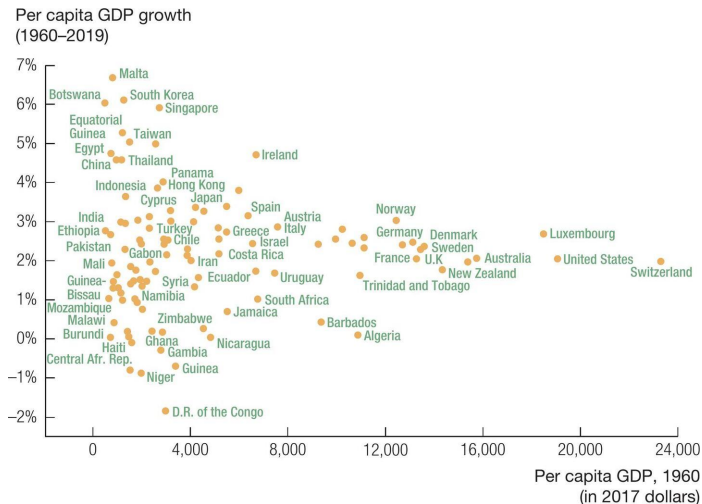
- In this case, countries that are poor in 1960 would be far below their steady state
- Countries that are rich would be closer to (or even above) their steady state



Source: Penn World Table, Version 10.0.

Growth Rates around the World, 1960–2017

- On average, rich and poor countries grow at the same rate
 - But, the variance is high for countries with low initial GDP



Case Study: South Korea and the Philippines

- South Korea
 - ▶ Grow at 6.4 percent per year
 - ▶ Increased from 10 percent of U.S. income to 75 percent
- Philippines
 - ▶ Grow 2.5 percent per year
 - ▶ Stayed around 10 percent of U.S. income
- Transition dynamics predicts that
 - ▶ South Korea was far below its steady state – high growth rate
 - ▶ Philippines is already at steady state – low growth rate

Case Study: South Korea and the Philippines

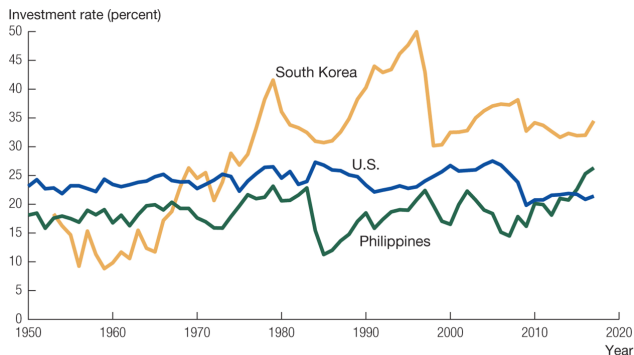
- Assuming equal depreciation rates

$$\frac{Y_{KR}^*}{Y_{US}^*} = \left(\frac{\bar{A}_{KR}}{\bar{A}_{US}} \right)^{3/2} \left(\frac{\bar{s}_{KR}}{\bar{s}_{US}} \right)^{1/2}$$

- The long-run ratio of per capita incomes depends on
 - ▶ the ratio of productivities (TFP levels)
 - ▶ the ratio of investment rates

Investment in South Korea and the Philippines, 1950–2017

- The investment rate in South Korea rose sharply between 1960 and 1990, from about 15 percent in the 1950s to around 40 percent in the early 1990s
- The steady state income in Korea should increase according to the Solow model
 - ▶ In order to transition to its new steady state, Korea grew at a faster rate
- Investment rates remained relatively stable in the United States and the Philippines



Source: Penn World Tables, Version 9.1.
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Strengths and Weaknesses of the Solow Model

- A key element of the Solow model is a production function that
 - ▶ depends on capital and labor and an accumulation equation
 - ▶ shows how forgoing consumption today leads to a higher capital stock tomorrow
- The strengths of the Solow model:
 - ▶ A baseline model that is able to determine how rich a country is in the long run (Steady State)
 - ★ Rich countries have a high rate of investment, a high TFP level, and/or a low rate of depreciation
 - ▶ The transition dynamics allows for an understanding of differences in growth rates across countries
 - ▶ A country further from (closer to) the steady state will grow faster (slower)
 - ★ Matching the convergence pattern in the data for some countries

Strengths and Weaknesses of the Solow Model

- The weaknesses of the Solow model:
 - ▶ differences in investment rates explain only a small fraction of the differences in income across countries
 - ▶ The much more important factor of TFP is still unexplained
- It does not explain why different countries have different investment and productivity rates
 - ▶ Economists extending the Solow model have endogenized the investment rate
 - ▶ e.g., Korean government removed substantial barriers to investment, and the Philippines maintained these barriers
- The model does not provide a theory of sustained long-run economic growth
 - ▶ Diminishing returns to capital accumulation in the production function mean that capital accumulation by itself cannot sustain growth

Next Lecture

- Next lecture will start Ch.6